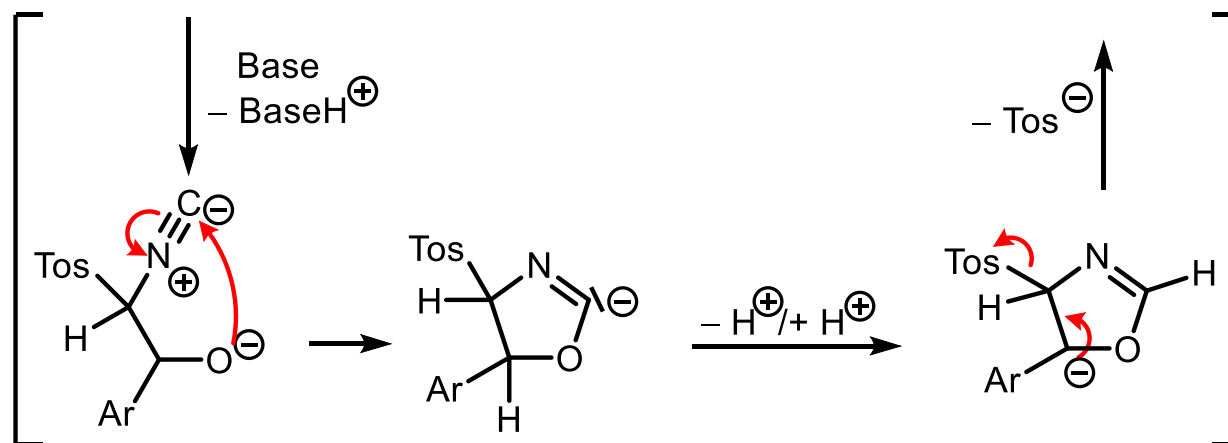
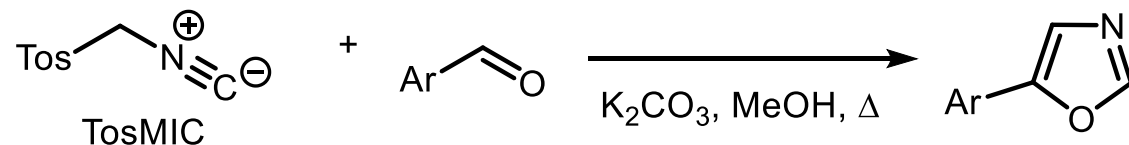
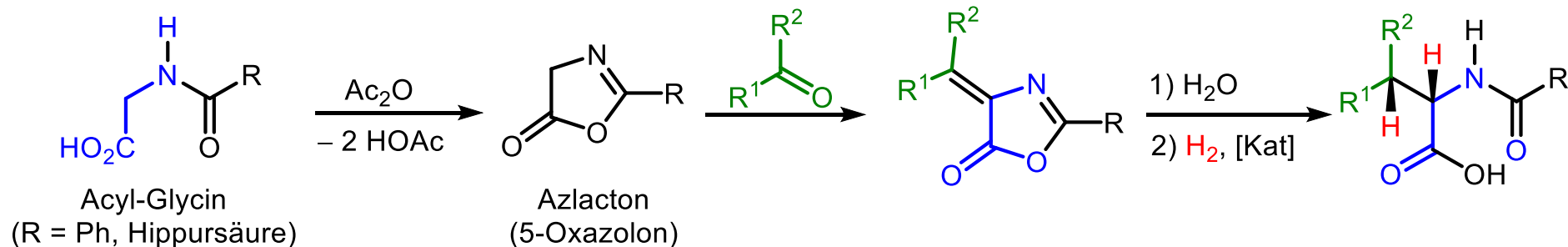
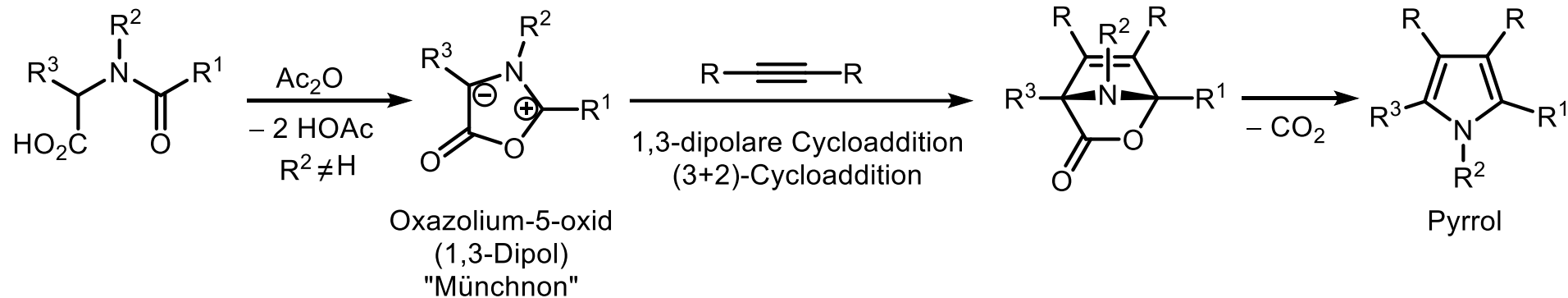


b) van Leusen-Synthese (1972):**c) Erlenmeyer-Azlacton-Synthese: guter Zugang zu Aminosäuren**

4.6.3. Ringaufbauende Synthesen

Fortsetzung



Rolf Huisgen
(1920-2020)

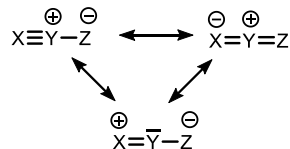
4.6.4. 1,3-Dipolare Cycloadditionen

1,3-Dipole sind dreiatomige Systeme mit 4 π -Elektronen. Es können keine Resonanzstrukturen ohne „Ladungen“ formuliert werden.

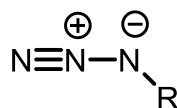
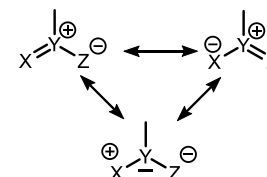
Reaktivität:

1,3-dipolare Cycloadditionen verlaufen für gewöhnlich „konzertiert“, d.h. ohne erkennbare Zwischenstufe. Der cyclische Übergangszustand bildet sich durch suprafacial-suprafaciale Anlagerung der beiden Komponenten (Dipol, Dipolarophil, Olefin oder Alkin).

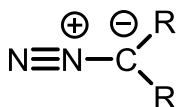
Propargyl-Allenyl-Typ



Allyl-Typ



Azide



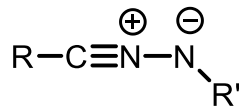
Diazoverbindungen



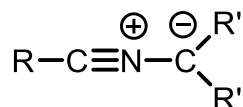
Nitriloxide



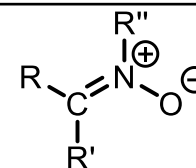
Nitrilsulfide



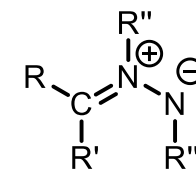
Nitrilimide



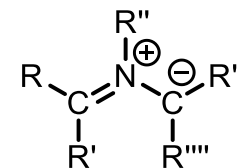
Nitrilylide



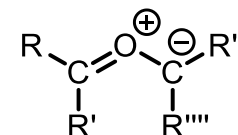
Nitrone



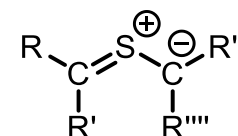
Azomethinimide



Azomethinylyde



Carbonylylyde



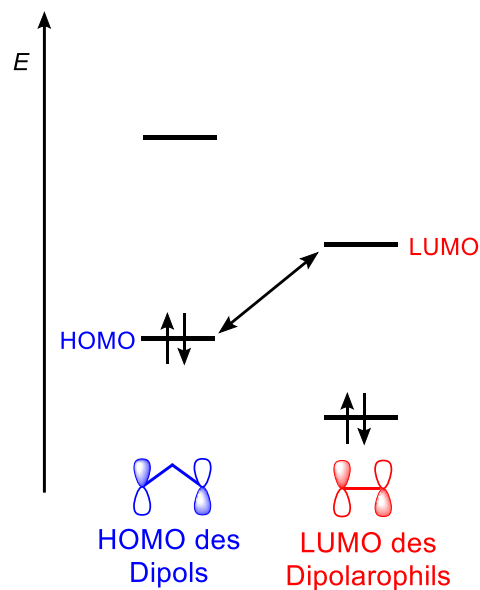
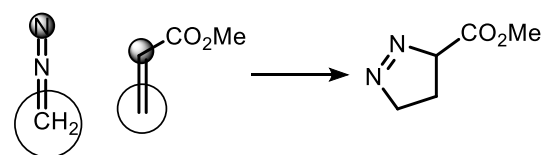
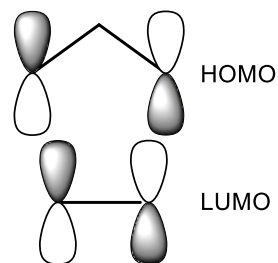
Thiocarbonylylyde

4.6.4. 1,3-Dipolare Cycloadditionen

Fortsetzung

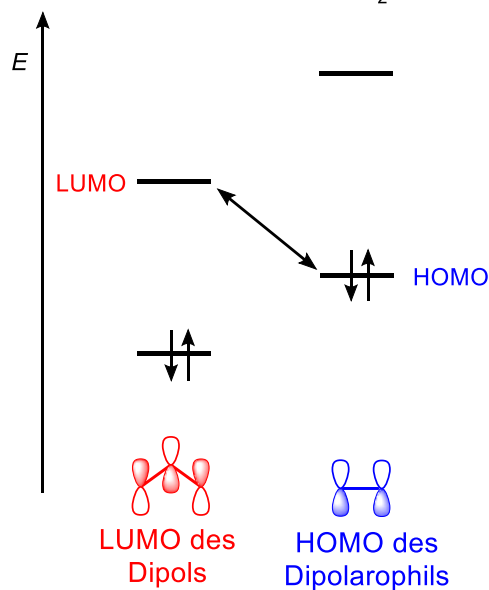
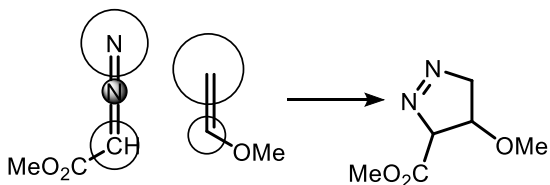
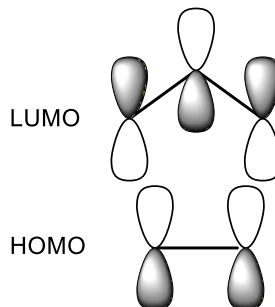
Grenzorbitalbetrachtung

normaler Elektronenbedarf

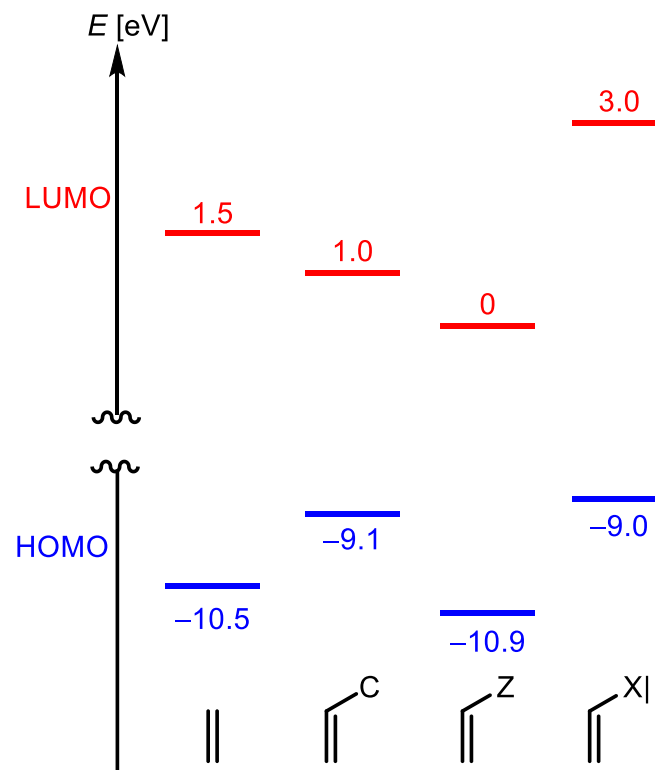


Heterocyclen

inverser Elektronenbedarf

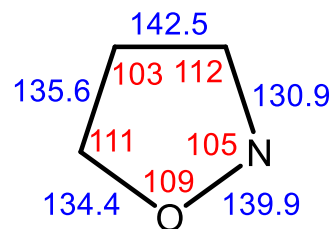
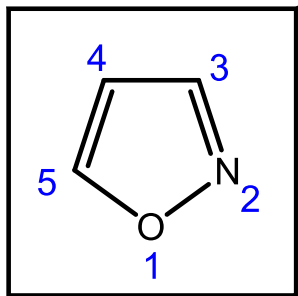


Prof. Dr. T. J. J. Müller

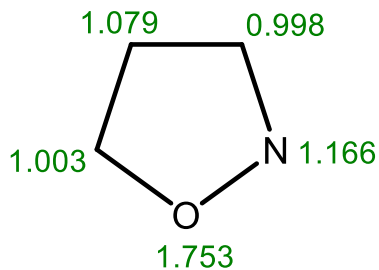


C: Vinyl, Phenyl
Z: CHO, CN, NO₂
X|: MeO, Me₂N, Me

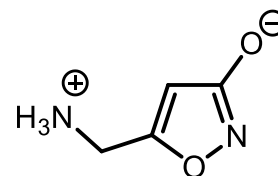
4.6.5. Isoxazol



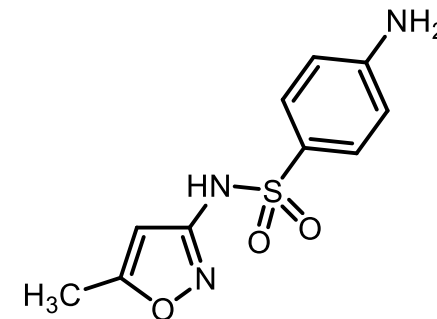
Bindungslängen [pm]
Bindungswinkel [°]



π -Elektronendichte



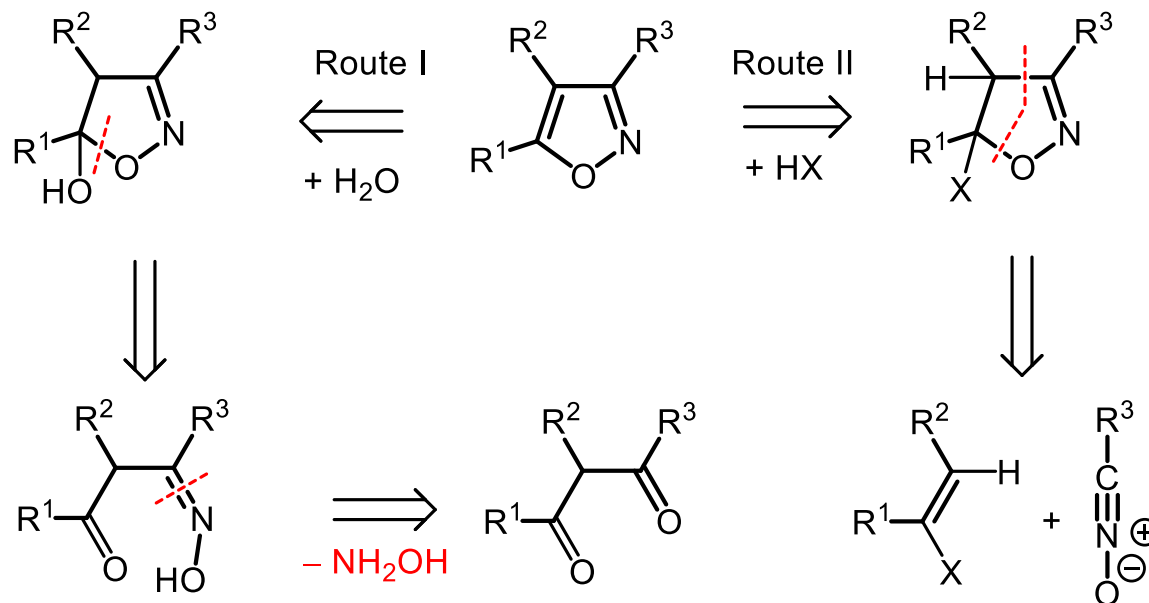
Muscimol
(Fliegenpilz)
psychoaktiv



Sulfamethoxazol
(Langzeitsulfonamid-
Antibiotikum bei Harnwegs-
infektionen)

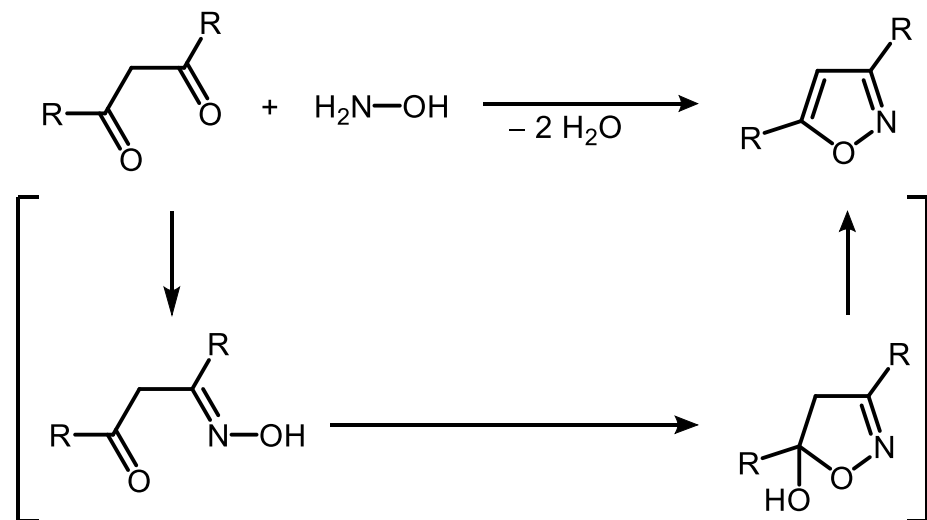
Retrosynthetische Analyse

66

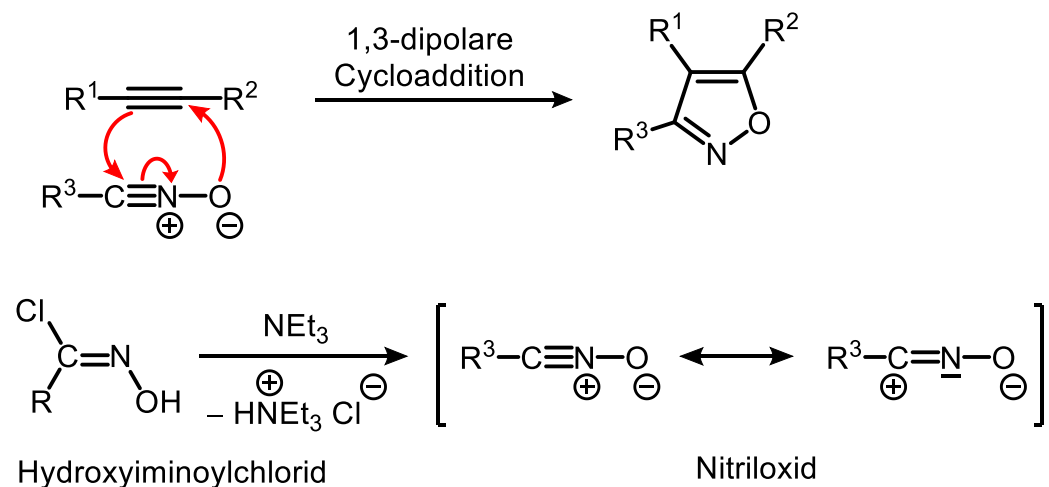


4.6.5. Isoxazol Fortsetzung

1) Claisen-Synthese

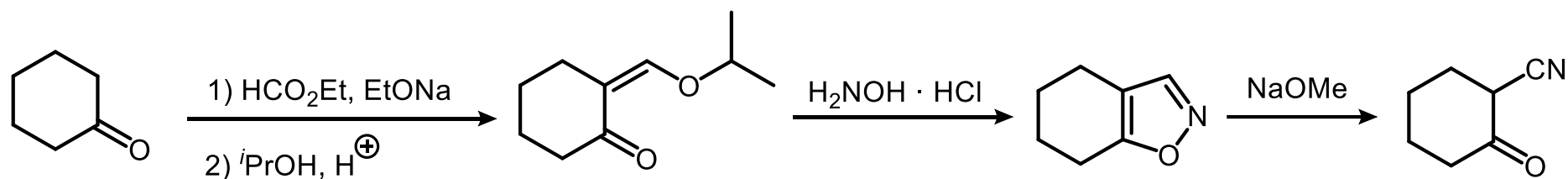
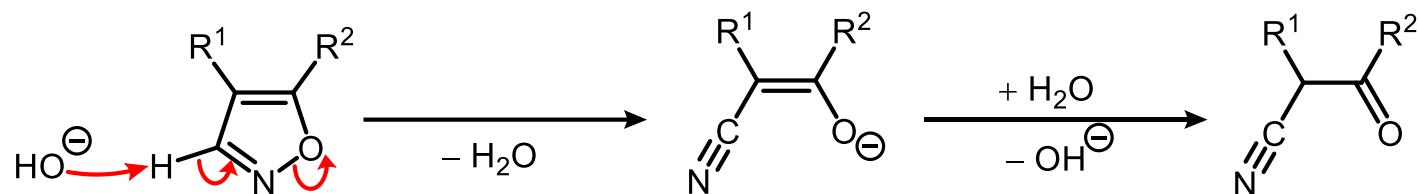


2) Quilico-Synthese

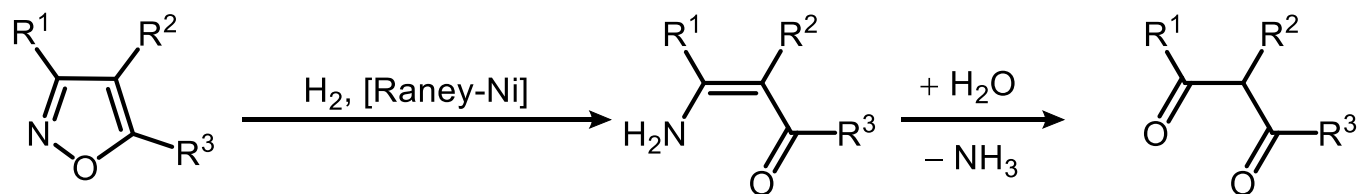


4.6.5. Isoxazol Fortsetzung

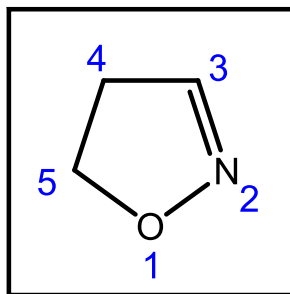
3) Reaktion mit Nucleophilen



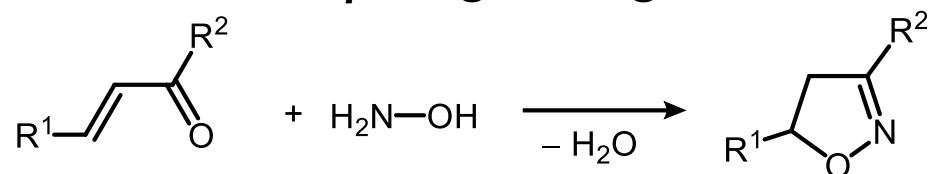
4) Reduktive Ringöffnung



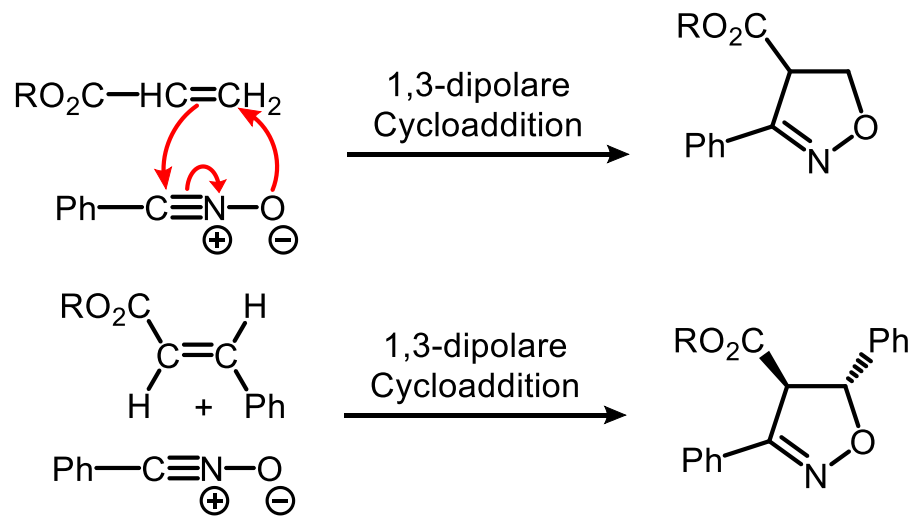
4.6.6. 4,5-Dihydroisoxazol



a) Cyclokondensation von α,β -ungesättigten Ketonen mit Hydroxylamin:

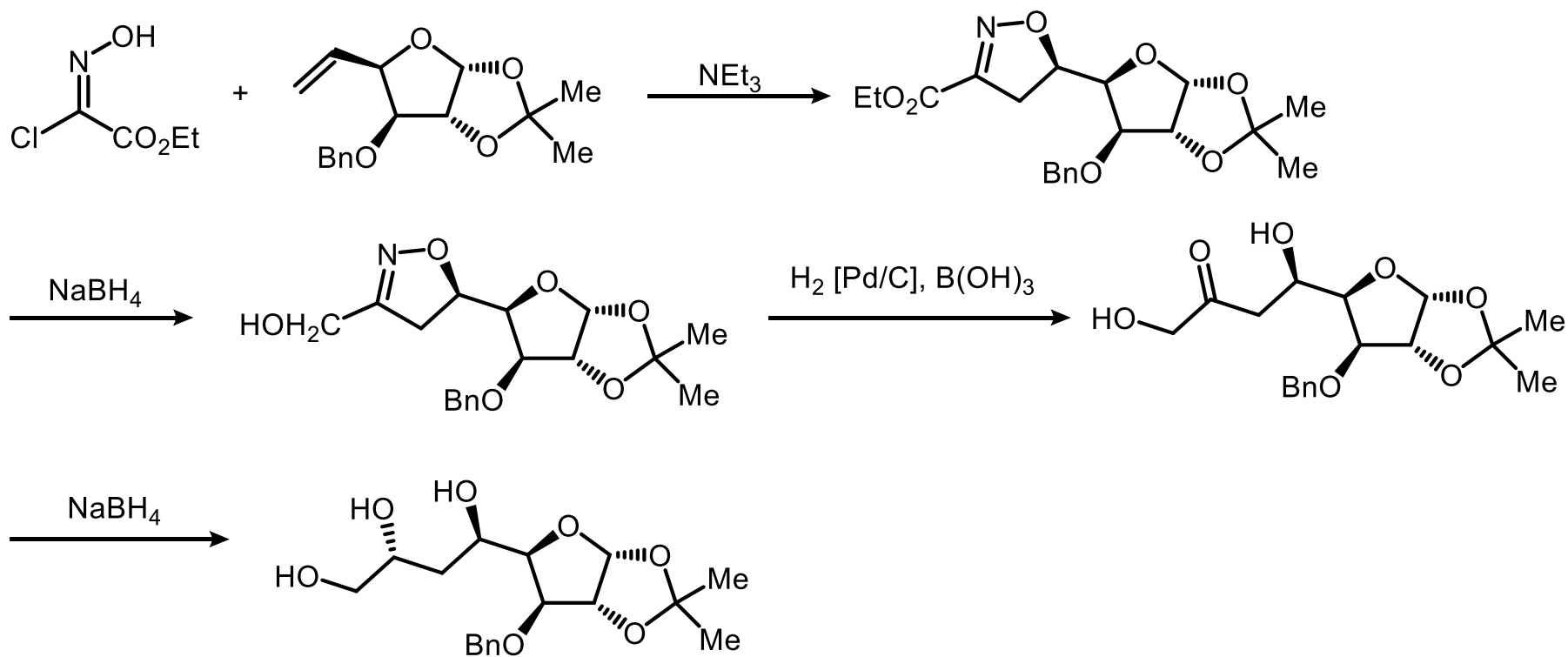
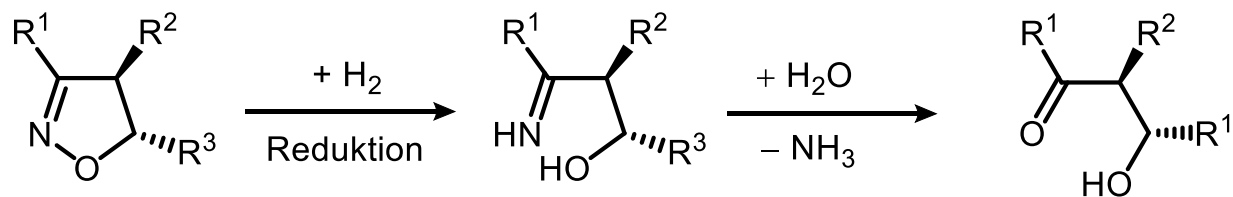


b) 1,3-dipolare Cycloaddition von Nitriloxiden an Alkene:



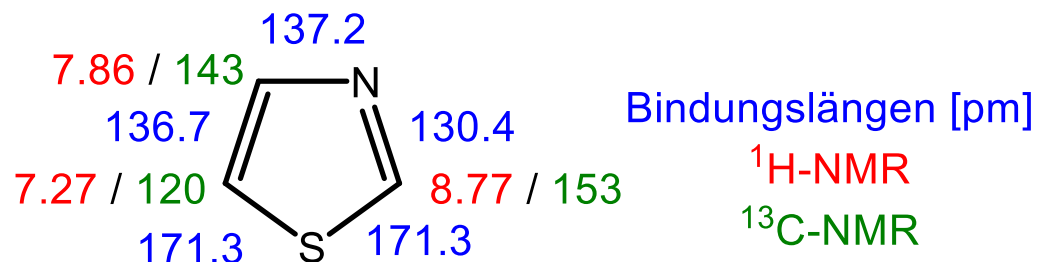
4.6.6. 4,5-Dihydroisoxazol

Fortsetzung



4.7. Thiazol

4.7.1. Struktur, Eigenschaften



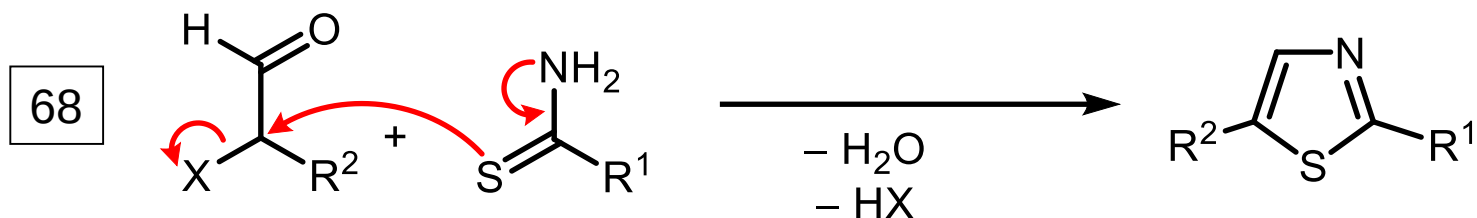
Sdp. 118 °C

4.7.2. Reaktivität

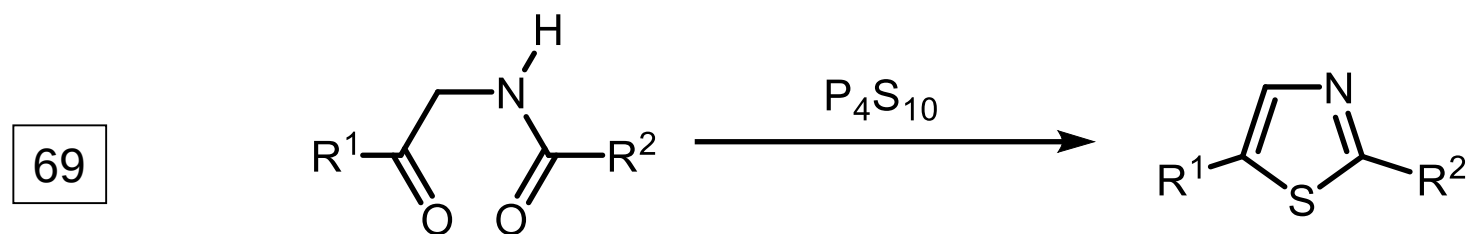
- elektronenärmste Position C₂.
- 67 - Lithiierung an C₂ (+ DMF → 2-Formylierung).
- Quaternierung mit Haloalkanen am N.
- *keine Diels-Alder-Reaktionen*

4.7.3. Ringaufbauende Synthesen

a) Hantzsch-Synthese

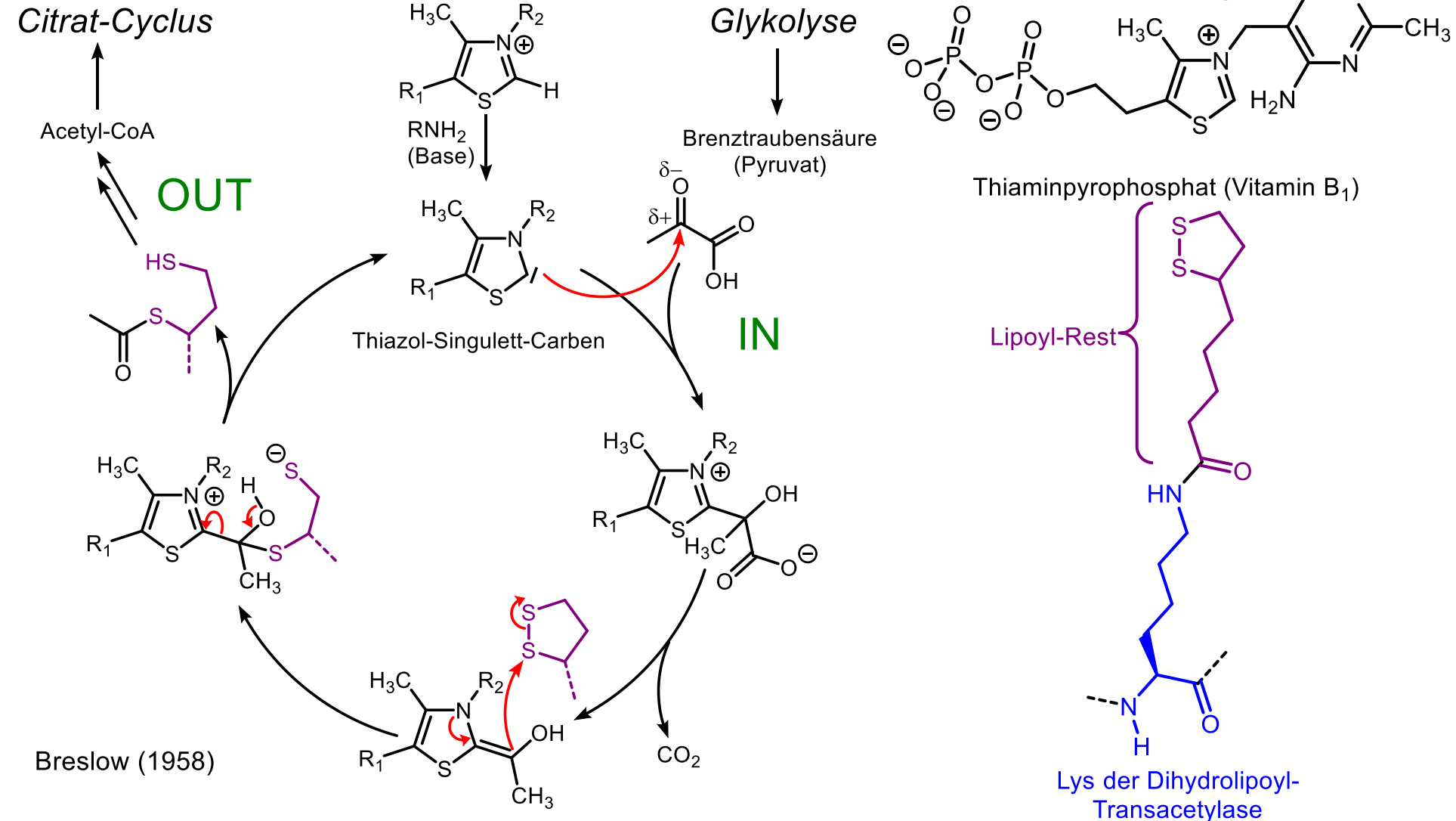


b) Gabriel-Synthese



4.7.4. Biologische Bedeutung

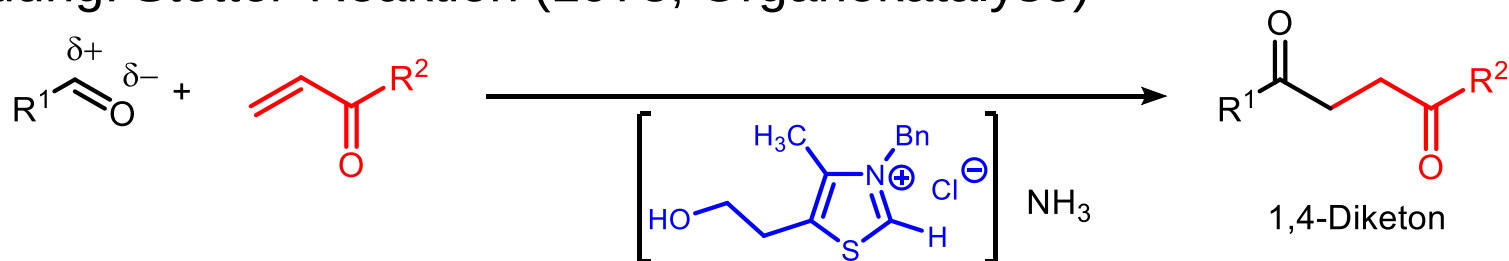
Oxidative Decarboxylierung von Pyruvat



4.7.5. Anwendung

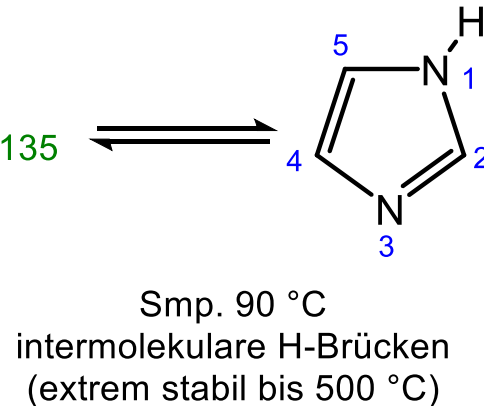
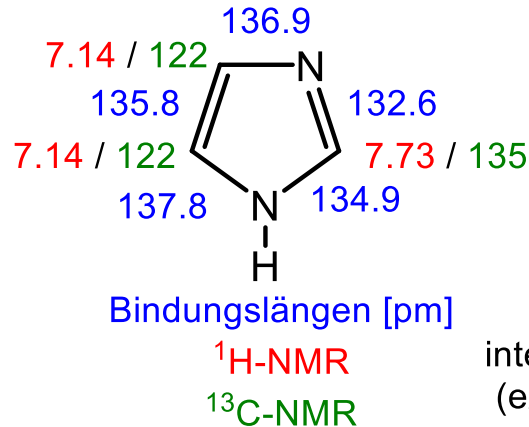
Anwendung: Stetter-Reaktion (1973, Organokatalyse)

70

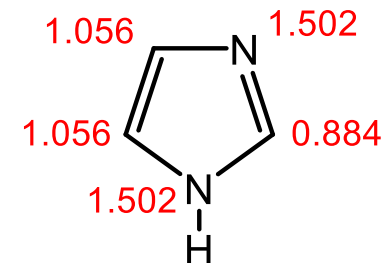


4.8. Imidazol

4.8.1. Struktur, Eigenschaften

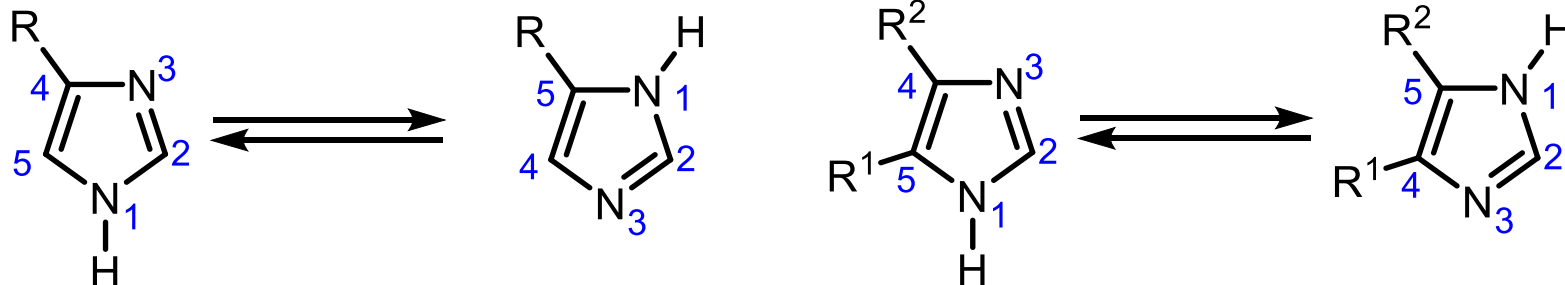


π -Elektronendichteverteilung



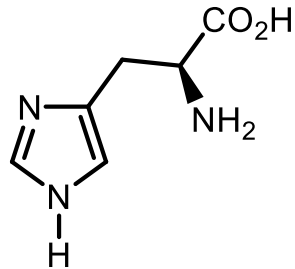
- pK_a (ImH₂⁺): 7.00, pK_a (ImH): 14.52 (N-Deprotonierung mit NaOEt).
- annulare Tautomerie(!): 4-Nitroimidazol, 5-Methoxyimidazol.

71

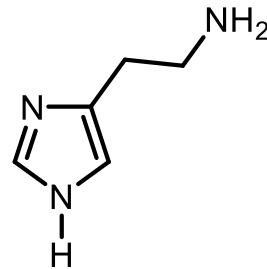


4.8.2. Vorkommen und Bedeutung

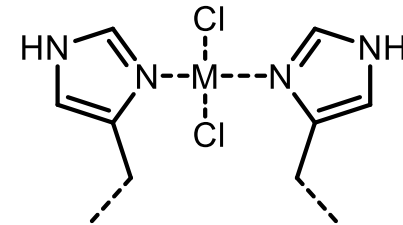
Naturstoffe



Histidin

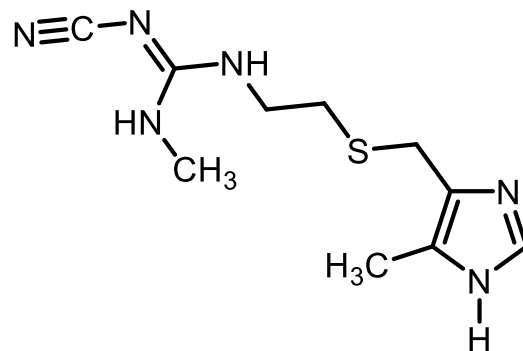


Histamin

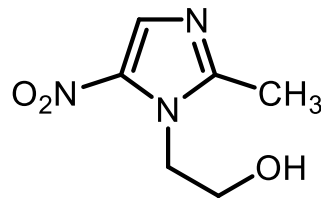


Metallkomplexe (Häm (Fe), Vitamin B₁₂ (Co), ...)

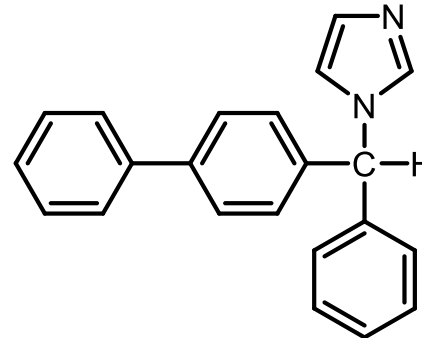
Pharmaka



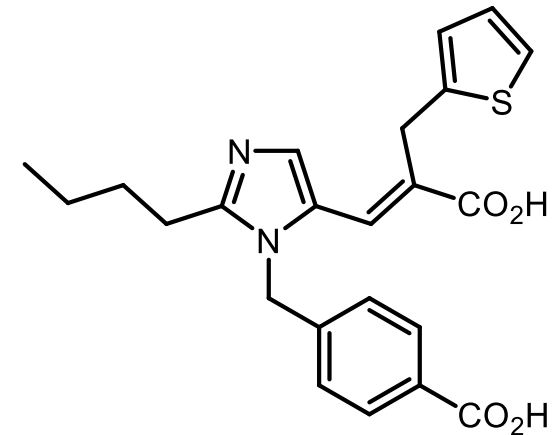
Cimetidin
(H₂-Rezeptor-Blocker)



Metronidazol
(Antiinfektivum)



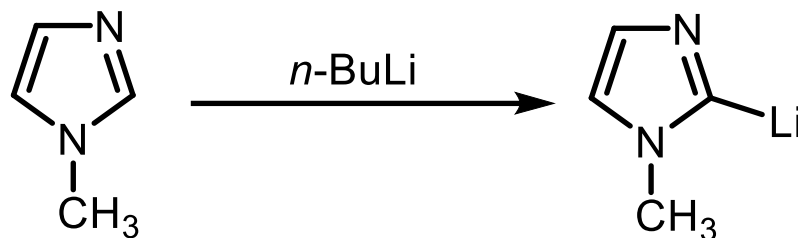
Bifonazol
(Antimycoticum)



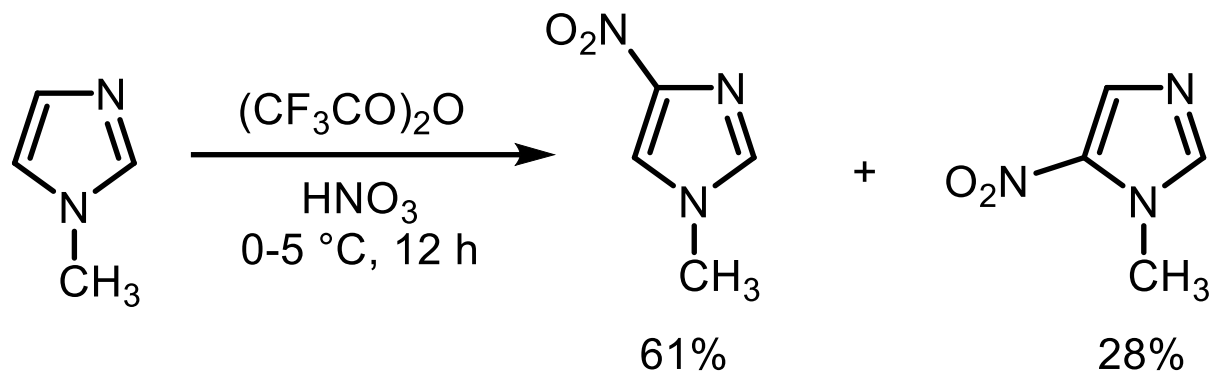
Eprosartan
(Antihypertensivum)

4.8.3. Reaktivität

- Lithiierung an C₂ (bei 1-subst. Imidazolen).
- Lithiierung an C₅ (bei 1,2-disubst. Imidazolen).



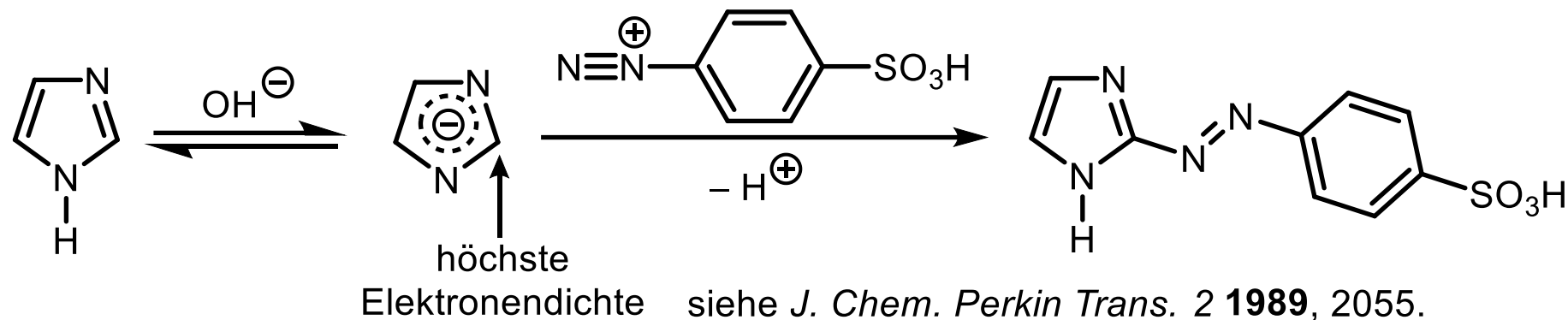
S_EAr : N-Elektrophile; parallel: reversibler elektrophiler Angriff an den N-Positionen
Nitrierung der 4- und 5-Position (saure Bed., *langsam*, ebenso wie 4-Sulfonierung)
größte C-Elektronendichte beim Imidazolium-Ion ImH_2^+ in Pos. 4 und 5.



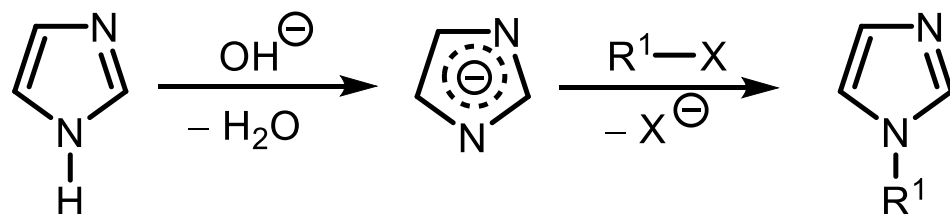
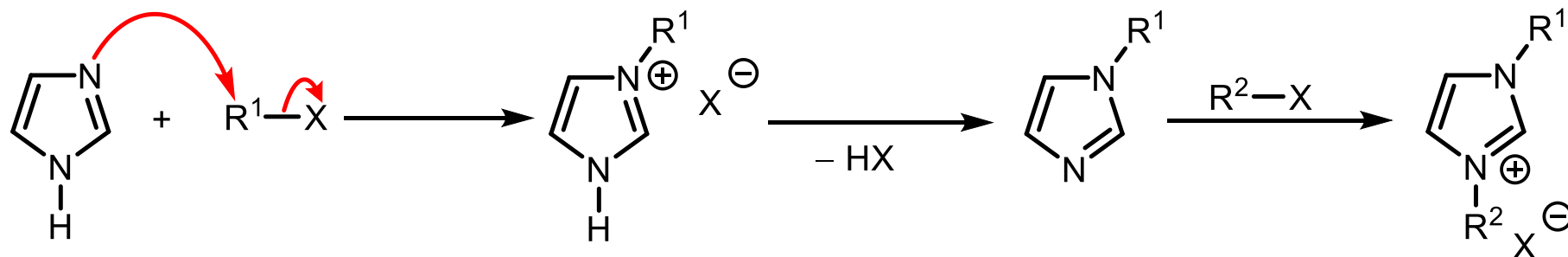
4.8.3. Reaktivität

Fortsetzung

aber: leichte Diazotierung der 2-Position (schwach alkal. Bed.)



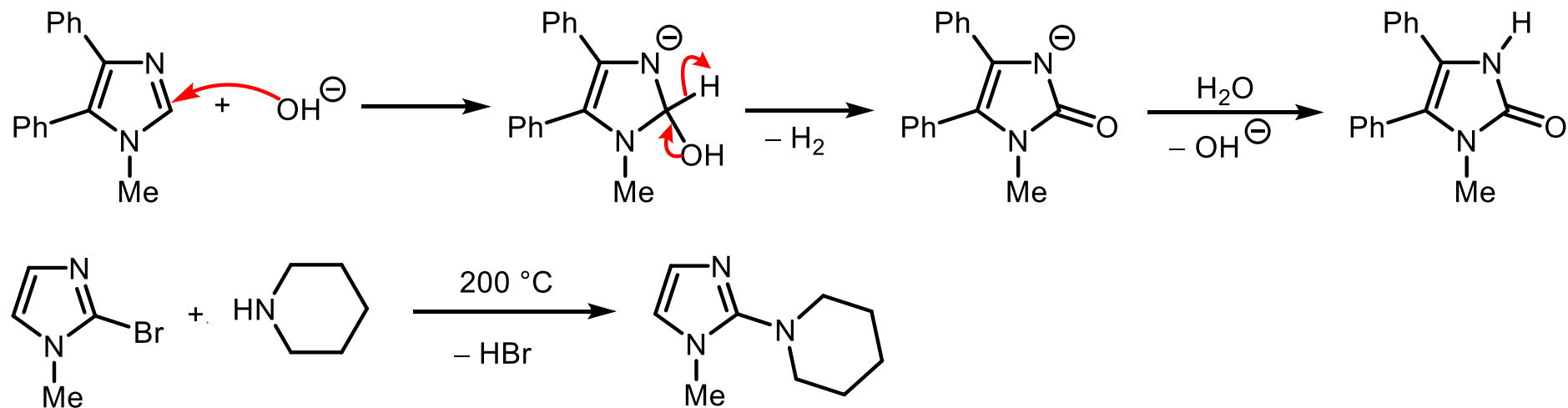
Reaktionen mit Elektrophilen



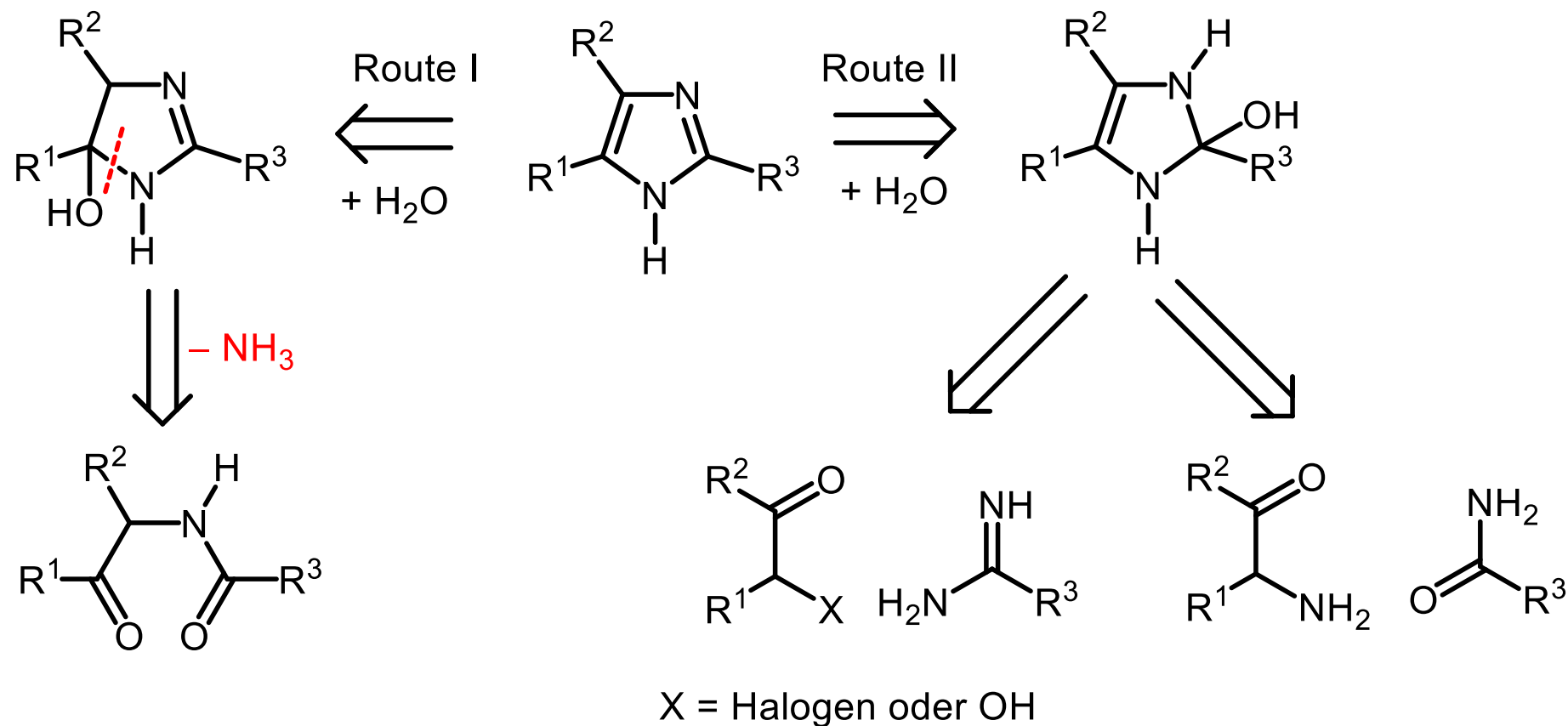
4.8.3. Reaktivität

Fortsetzung

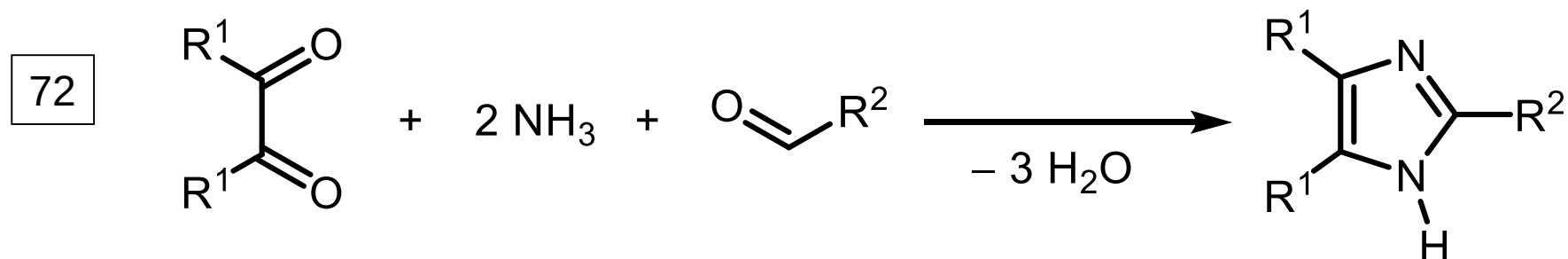
Reaktionen mit Nucleophilen



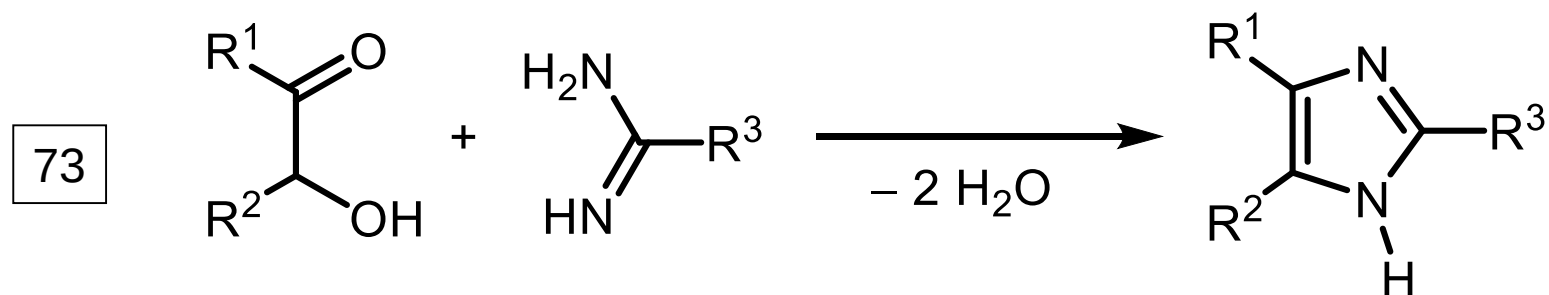
4.8.4. Ringaufbauende Synthesen



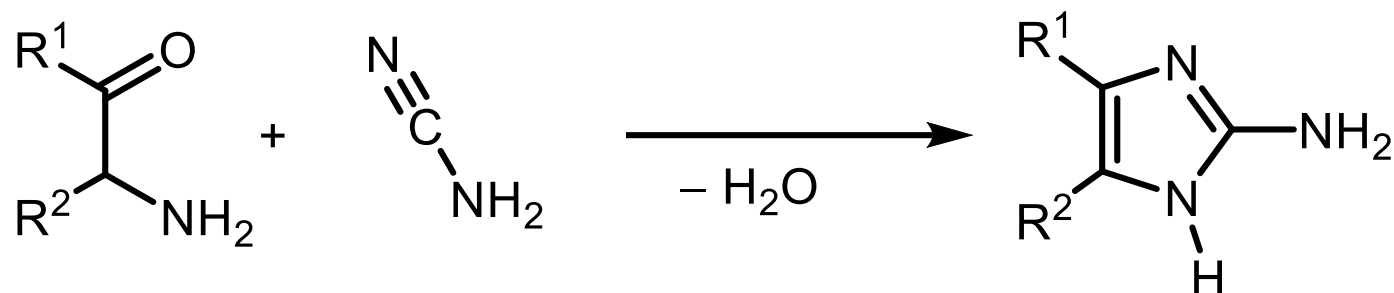
1) Cyclokondensation von 1,2-Dicarbonyl-Verb. / Ammoniak / Aldehyden (Debus-Radziszewski-Synthese)



2) aus α -Halogen- oder α -Hydroxyketonen und Amidinen

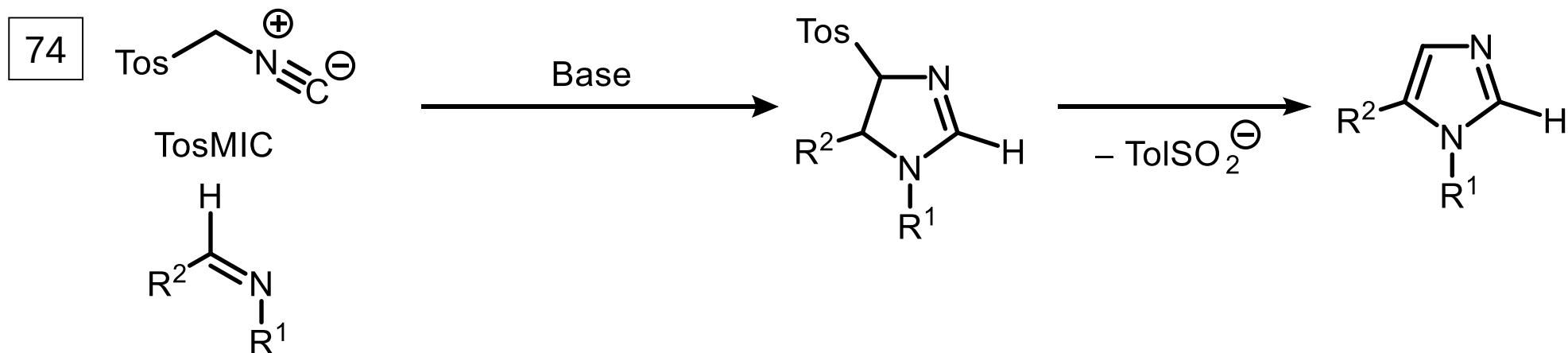


3) Marckwald-Synthese



4) van-Leusen-Synthese

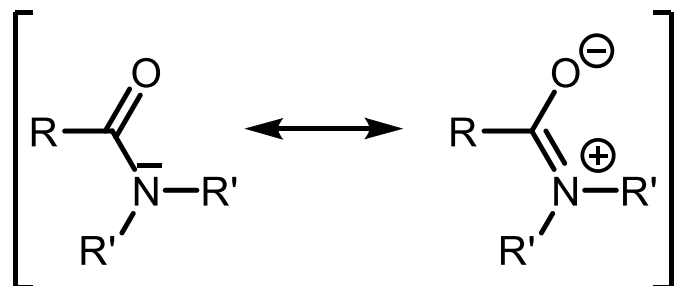
aus Aldiminen und Tosylmethylenisocyanid → 1,5-disubstituierte Imidazole



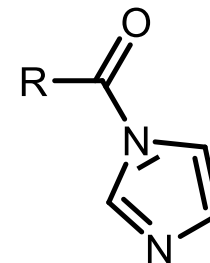
4.8.5. Imidazolreagenzien

a) Imidazol als Acylgruppen-Überträger

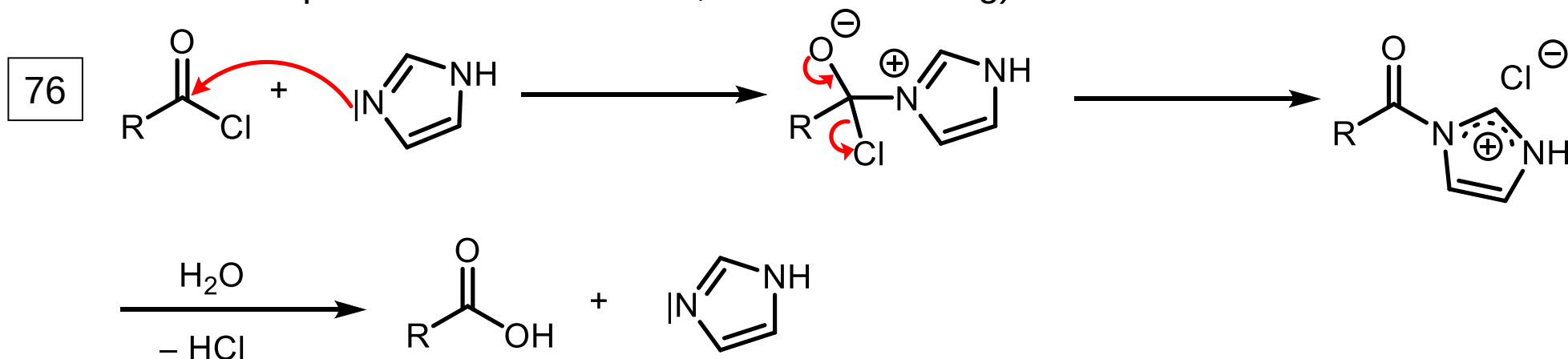
Übliche Mesomerie-Stabilisierung von Amiden



Keine Mesomerie-Stabilisierung, da das Elektronenpaar in die 6 π -Resonanz eingebettet ist

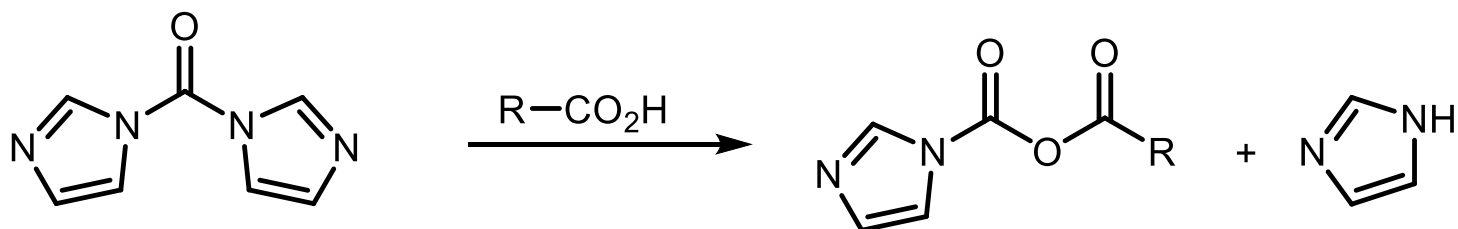
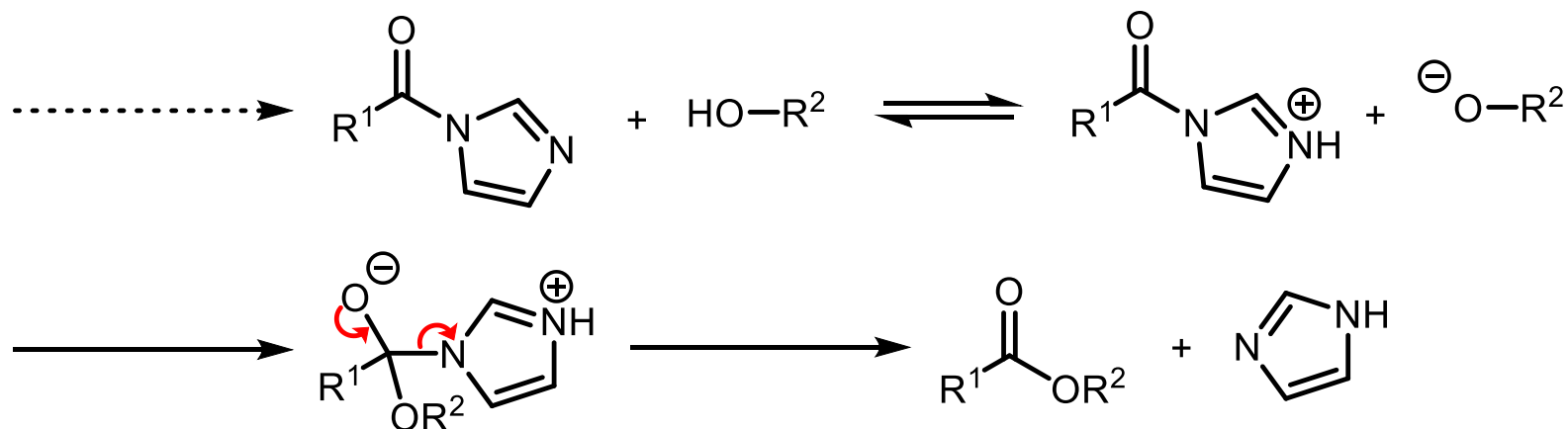


Nukleophile Katalyse (hier Hydrolyse von Säurechloriden; ebenso bei Alkoholyse mit sterisch anspruchsvollen Alkoholen, d.h. Veresterung):

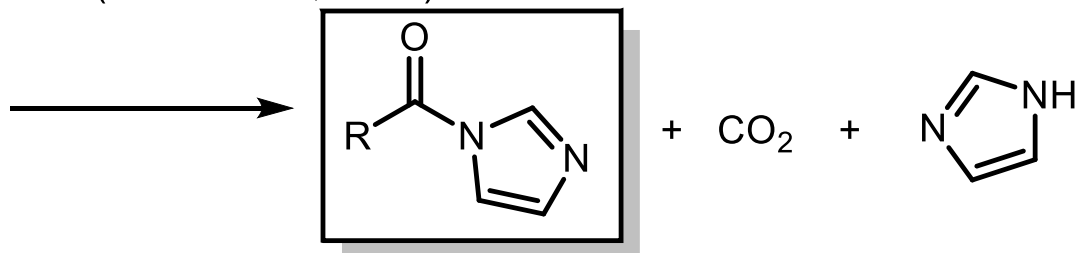


4.8.5. Imidazolreagenzien

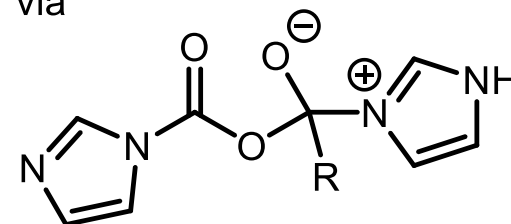
Fortsetzung



1,1'-Carbonyldiimidazol (CDI)
Staabs Reagenz
(H. A. Staab, 1956)



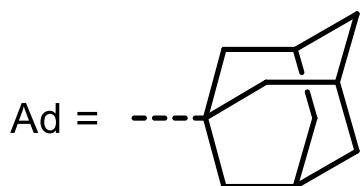
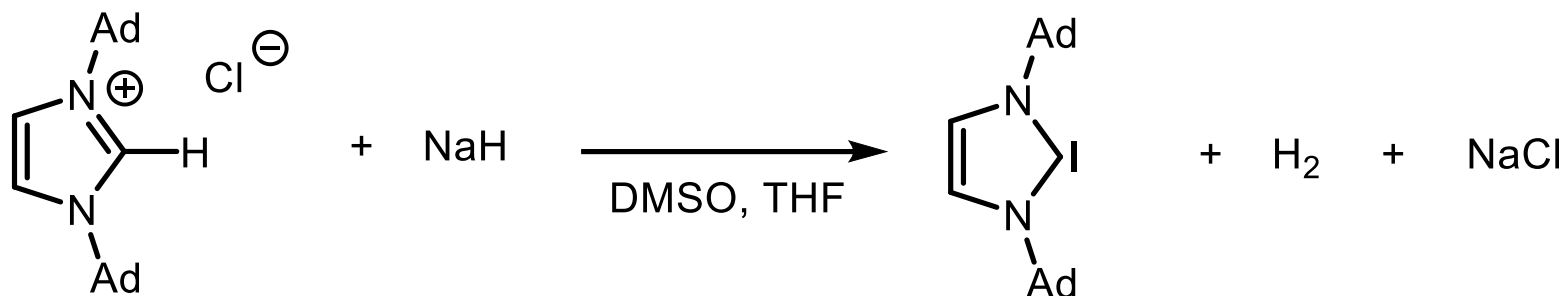
via



4.8.5. Imidazolreagenzien

Fortsetzung

b) Imidazol-Carbene (2-Deprotonierung von N,N'-Dialkyl-(bzw. -aryl)imidazolium-Ionen: stabile Carbene!)



Singulett-Carben (formal leeres p_z -Orbital, doppelt besetztes sp^2 -Orbital); stabilisiert durch die Delokalisierung der benachbarten N-p-Elektronenpaare
 A. J. Arduengo III, *J. Am. Chem. Soc.* **1991**, 361; *Acc. Chem. Res.* **1999**, 913.

„Grubbs-Katalysatoren“ zur Alken- und Enin-Metathese (Nobelpreis für Chemie 2005)
 R. H. Grubbs, *Angew. Chem.* **2006**, 118, 3845.

